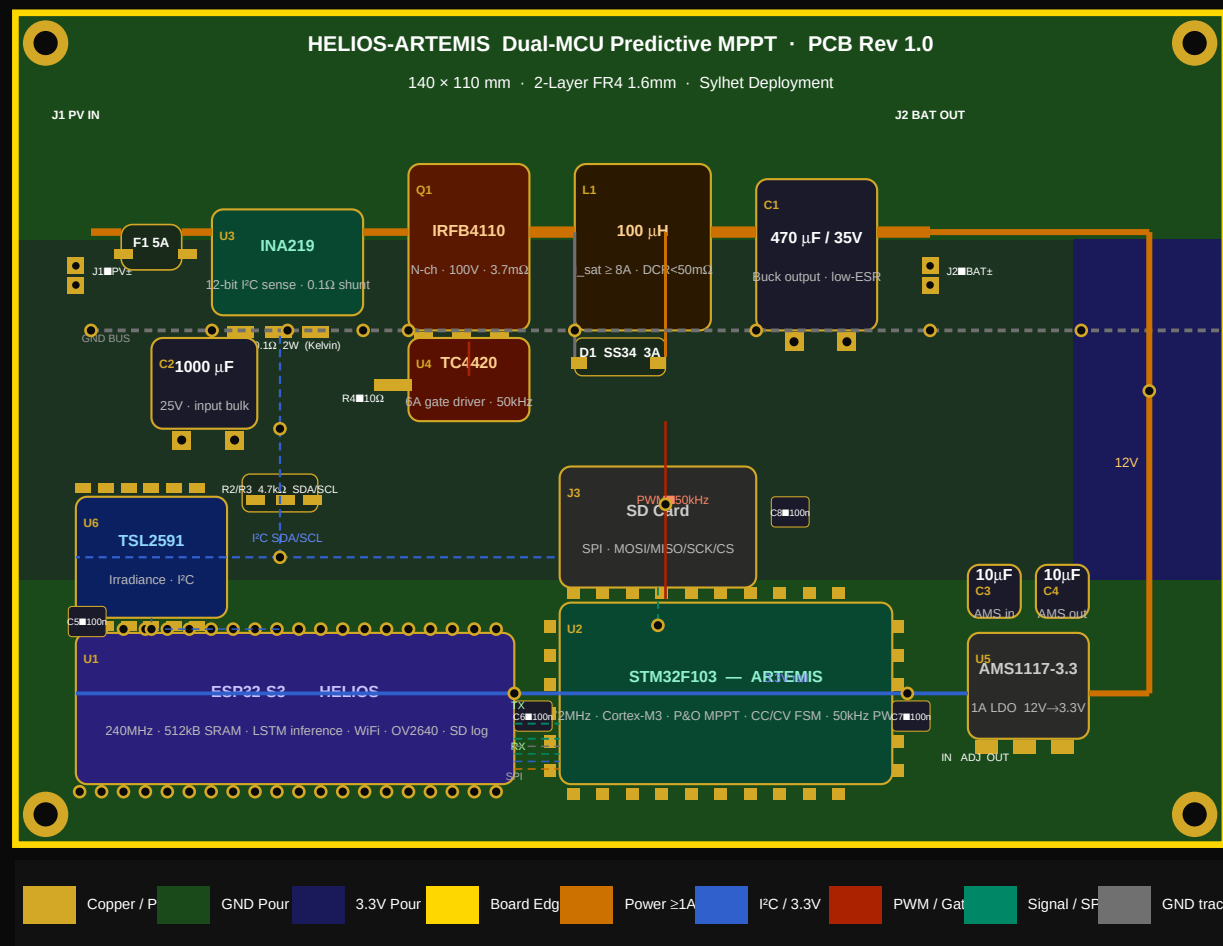


HELIOS-ARTEMIS Dual-MCU Predictive MPPT — PCB Layout Rev 1.0 · 140x110mm 2-Layer FR4



HELIOS-ARTEMIS MPPT — Bill of Materials & Assembly Notes

Ref	Part / Value	Description	Specifications	Notes	BDT
U1	ESP32-S3 DevKit-C	MCU — Helios (AI)	240MHz · 512kB SRAM · WiFi · Dual-core	380	
U2	STM32F103C8T6	MCU — Artemis (ctrl)	72MHz Cortex-M3 · LQFP48	120	
U3	INA219 breakout	Current / voltage sense	12-bit I²C · 0.1Ω shunt · addr 0x40	80	
U4	TC4420 CPA	MOSFET gate driver	6A peak · 12V supply · SO-8	—	
U5	AMS1117-3.3	3.3V LDO regulator	1A output · SOT-223 · heatsink pad	30	
U6	TSL2591	Ambient light sensor	I²C · 600 Mlux dynamic range	120	
Q1	IRFB4110 (TO-220)	N-channel power MOSFET	VDS=100V · ID=120A · RDS(on)=3.7mΩ	—	
D1	SS34 Schottky	Freewheeling diode	3A · 40V · DO-214AB (SMA)	—	
L1	100 μH power inductor	Buck filter inductor	I_sat ≥ 8A · DCR < 50mΩ · radial TH	—	
C1	470 μF / 35V electrolytic	Buck output filter	Low-ESR · 105°C · radial TH	—	
C2	1000 μF / 25V electrolytic	Input bulk capacitor	Place close to Q1 drain	—	
C3,C4	10 μF tantalum	AMS1117 in/out caps	Required for LDO stability	—	
C5–C8	100 nF / 0603 MLCC	Bypass capacitors	One at each IC VCC pin · close to pin	—	
R1	0.1 Ω / 2W resistor	INA219 shunt	4-wire Kelvin connection in power path	—	
R2,R3	4.7 kΩ / 0603	I²C pull-up resistors	SDA + SCL lines to 3.3V rail	—	
R4	10 Ω / 0603	Gate series resistor	Limits dV/dt · place near TC4420 OUT pin	—	
J1	2-pin screw terminal	PV panel input	5A rated · 5mm pitch	—	
J2	2-pin screw terminal	Battery output	10A rated · 5mm pitch	—	
J3	SD card module (SPI)	Data logging	MOSI/MISO/SCK/CS · 3.3V logic	—	
—	PCB fabrication	2-layer FR4 140×110mm	HASL · green mask · white silk	—	
—	Misc passives + housing	Connectors, wire, enclosure	Includes TC4420 + diode cost	630	
TOTAL CONTROLLER BOM				≈ 1,360 BDT	(89% below commercial MPPT price)

ASSEMBLY & ORDERING NOTES

- Power path: Keep J1→F1→R1→Q1→L1→J2 traces ≥1.5mm wide. Minimise loop area of Q1 drain-source path. Place C2 (1000μF) within 10mm of Q1 drain.
- Gate driver: TC4420 needs 12V supply with 100nF bypass at Vcc pin. Route PWM trace (STM32 PA8 → TC4420 IN) away from I²C and SPI signals. Place R4 (10Ω) at TC4420 output.

Helios-Artemis MPPT · BOM Rev 1.0 · Sylfhet Deployment 2026

Total BOM: ≈ 1,360 BDT

I²C bus: R2/R3 pull-ups (4.7kΩ) placed near ESP32 GPIO 21/22. Shared bus: INA219 addr=0x40. TSL2591 addr=0x29. STM32 reads INA219 on PB6/PB7 (same bus).

Layer Stack-up (2-Layer FR4 1.6mm)

L1 — Top Copper (35μm):	Signal routing, SMD pads, power traces, component footprints
Prepreg:	FR4 core 1.6mm total finished thickness
L2 — Bottom Copper (35μm):	GND plane flood-fill, return current paths, thermal relief
Surface finish:	HASL (Hot Air Solder Levelling) — lead-free
Solder mask:	Green LPI (Liquid Photo-Imageable), both sides
Silkscreen:	White epoxy ink, top side only

Design Rules

Minimum trace width:	0.2mm (signal) · 1.0mm (logic power) · 1.5mm (PV/battery path)
Minimum clearance:	0.2mm (signal–signal) · 0.5mm (HV to logic) · 1.0mm (board edge)
Minimum drill diameter:	0.3mm (via) · 0.9mm (TH component) · 3.2mm (M3 mounting)
Minimum annular ring:	0.25mm (via) · 0.4mm (TH pad)
Via drill / pad:	0.6mm drill / 1.2mm pad (signal via) · 1.0mm drill / 1.8mm pad (power via)
IPC Class:	IPC-2221A Class 2 (general electronic products)

Critical Layout Rules

POWER LOOP: Keep Q1 drain → L1 → output path as a short, wide, low-inductance loop. Place C2 (1000μF) within 10mm of Q1 drain pin.
GATE DRIVE: Place R4 (10Ω gate resistor) physically at TC4420 OUT pin. Route gate trace directly to Q1 gate — no vias in this path.
INA219 SHUNT: Use 4-wire Kelvin connection for R1. Sense lines (SDA/SCL analogue inputs) must not carry switching current.
I²C ROUTING: Keep SDA/SCL away from the 50kHz PWM trace and switching node. Add ground guard trace if necessary.
THERMAL: Place thermal via arrays (4×4 grid, 0.6mm drill) under Q1 source pad and AMS1117 tab. Connect to GND plane.
TSL2591: Mount near board edge with unobstructed view of sky. Shield from direct IR from Q1/L1 heat sources.
DECOUPLING: 100nF bypass cap at every IC VCC pin. Place within 2mm of pin. Short return path to GND via.

Test & Programming Sequence

Step 1: Power 3.3V rail only (disconnect 12V). Measure: +3.30V ±0.05V at ESP32 VCC and STM32 VCC.
Step 2: Run I²C scan. Expected devices: 0x29 (TSL2591), 0x40 (INA219). Both must respond.
Step 3: Program STM32F103 via SWD (use ST-Link V2). Flash Artemis firmware. Verify 50kHz PWM on PA8 with oscilloscope.
Step 4: Program ESP32-S3 via USB-UART (DevKit-C onboard). Flash Helios firmware. Verify WiFi connection and I²C reads.
Step 5: Connect 12V supply (no PV/battery). Verify TC4420 receives gate pulses. Verify AMS1117 stays cool.
Step 6: Connect PV panel with 1A current limit. Verify INA219 reads V and I. Verify buck output voltage.
Step 7: Full system test with 12V/7Ah SLA battery. Monitor CC→CV→Float transition via UART log.